

EZ Roomba Mobile Robot

Ultrasonic sensing and motor-shield integration

MOBILE ROBOTICS	ARDUINO	SENSORS
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Project overview

A documented mobile-robot project using the NewPing ultrasonic-sensor library and the Adafruit v1 motor-shield library. The repository notes the practical pin-allocation constraint created by the motor shield and recommends repurposing analog pins as digital I/O.

Engineering problem

A small autonomous platform must integrate ranging and motor control within limited microcontroller pins while keeping obstacle response predictable.

System architecture - Ultrasonic ranging supplies obstacle-distance measurements through NewPing. - An Adafruit v1 motor shield drives the mobile base. - Analog pins can be configured as digital I/O because the shield occupies many PWM-capable pins. - A simple behavior layer can map distance thresholds to forward, turn, and stop actions.	Engineering focus - Pin budgeting is treated as an early system-integration decision. - Sensor filtering and timeout handling prevent single bad readings from dominating motion. - Motor asymmetry should be calibrated rather than hidden in steering logic.
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Verification plan

- Measure stopping distance and avoidance success over repeated obstacle approaches.
- Test no-echo, close-range, angled-wall, and low-battery cases.
- Confirm wiring, library versions, and pin assignments in a reproducible setup record.

Status and evidence boundary Repository-documented project; the public README confirms its core libraries and pin-allocation guidance.
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